

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 2 (Generated practice paper - Set 16)**

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
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10	
11	
TOTAL	

Answer all questions in the spaces provided.

01

Table 1 shows the ecological footprint and biocapacity per person for four countries and the world in 2016.

Table 1

	Ecological footprint / gha per person	Biocapacity / gha per person	Human Development Index
USA	8.1	3.6	0.92
UK	4.4	1.4	0.92
China	3.6	1.0	0.75
Bangladesh	0.8	0.4	0.61
World	2.8	1.7	0.72

01

1

Define **ecological footprint** and **biocapacity**.

[2 marks]

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01

2

Explain what is meant by a **global hectare** and why it is used.

[2 marks]

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01

3

Use Table 1 to calculate how many Earths would be needed if everyone lived like the average US citizen.

[1 mark]

Answer

Earths

0 1 . 4

Use Table 1 to describe the relationship between development and ecological footprint, and explain the concept of Earth Overshoot Day.

[3 marks]

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0 1 . 5

Suggest **two** ways a country such as the UK could reduce its footprint without reducing its HDI.

[2 marks]

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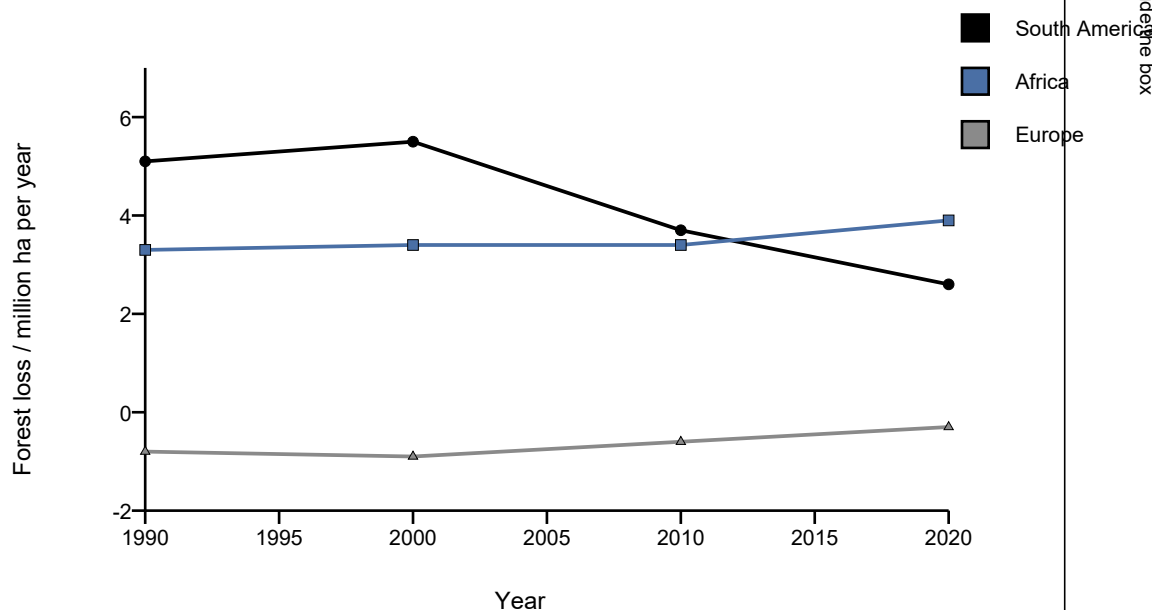
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10

0 2

Figure 1 shows the area of forest lost each year in three regions between 1990 and 2020.

Figure 1



0 2

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Describe the trends shown in Figure 1.

[3 marks]

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Explain the main causes of forest loss in South America and Africa.

[4 marks]

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0 2 . 3

Explain why roads and railways cause more forest loss than the area they occupy.

[2 marks]

0 2 . 4

Explain why softwood from Scandinavia and Canada is generally produced more sustainably than tropical hardwood.

[1 mark]

10

0 3

Table 1 shows some designated protected areas in the UK.

0 3 . 1

Complete Table 1 to show the legislation under which each area is designated and its key feature.

[4 marks]

Table 1

Designation	Designated under	Key feature	
Site of Special Scientific Interest (SSSI)			
Special Protection Area (SPA)			
Ramsar site			
Marine Conservation Zone (MCZ)			

03 . 2

Explain how designating a protected area protects habitats and species.

[3 marks]

03 . 3

Explain why voluntary agri-environmental schemes are needed in addition to legislation.

[3 marks]

10

04 . 1

Explain the difference between in-situ and ex-situ conservation and state when ex-situ conservation is needed.

[2 marks]

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04 . 2

Give **four** criteria used to select species for captive breeding and release programmes.

[4 marks]

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04 . 3

Explain why each of the following species would be difficult to keep or breed in captivity.

[4 marks]

Blue whale

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Large blue butterfly

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0 5

Table 1 shows the energy inputs and food energy outputs of three farming systems.

Table 1

System	Energy input / GJ ha⁻¹ yr⁻¹	Food energy output / GJ ha⁻¹ yr⁻¹
Extensive cereal farming (low inputs)	4	20
Intensive cereal farming (fertiliser, pesticide, machinery)	20	60
Intensive beef (grain-fed)	40	4

0 5**. 1**

Calculate the energy ratio (output / input) for each system.

[2 marks]**Answer**

0 5**. 2**

Use your answers to explain the difference between productivity and efficiency.

[3 marks]

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0 5**. 3**Explain what is meant by an **energy subsidy** and give **three** examples.**[3 marks]**

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- 0 5 . 4** Explain why spreading the same total inputs over a larger area would increase total food production.

[2 marks]

10

- 0 6** Table 1 compares extensive oyster farming with intensive salmon farming.

Table 1

Feature	Extensive oyster farm	Intensive salmon farm
Food input	None - filter natural plankton	Fishmeal / plant pellets
Stocking density	Low, spaced on seabed	High, in sea cages / tanks
Breeding	Wild spat settle on cleared seabed or hatchery	Selected adults stripped of eggs and milt
Disease / pest control	Removal of predatory crabs	Antibiotics, pesticides, wrasse, removal of diseased fish
Yield per hectare	Low	Very high

- 0 6 . 1** Use Table 1 to explain the relative merits of extensive and intensive aquaculture.

[4 marks]

0 6 . 2

Describe the stages in breeding and rearing salmon in an intensive farm.

[3 marks]

0 6 . 3

Explain how shrimp farming has affected mangroves and how disease is controlled in shrimp lagoons.

[3 marks]

10

0 7 . 1

Which of the following would be most suitable for monitoring the movements of a whale over several months? Tick one box.

[1 mark]

- A Radio collar
- B Acoustic / satellite tag
- C Pitfall trap
- D Geolocator tag recovered by recapture

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07 . 2

Explain why post-release monitoring is important in a captive breeding and release programme and describe **two** suitable methods.

[4 marks]

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08

A group of students wants to investigate the effect of vegetation cover on rain-splash soil erosion.

08 . 1

Describe how they could carry out the investigation in the laboratory.

[5 marks]

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08 . 2

Give **three** variables, other than those you have standardised, that affect rain-splash erosion in the field.

[3 marks]

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08 . 3

Explain why it is easier to obtain valid results in the laboratory than in the field, and give **one** disadvantage of laboratory studies.

[2 marks]

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09

Table 1 shows the numbers of breeding pairs of two seabird species on an island before and after a rat eradication programme.

Table 1

Species	Breeding pairs before eradication	Breeding pairs 8 years after eradication
Manx shearwater	45	310
Storm petrel	12	140

09 . 1

Explain why oceanic islands have many endemic species but low overall biodiversity.

[4 marks]

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09 . 2

Use Table 1 to explain why introduced rats had such a severe effect on these birds.

[3 marks]

09 . 3

Explain how avian malaria threatens Hawaiian birds and how climate change worsens the threat.

[2 marks]

09 . 4

Give **one** other conservation measure for oceanic islands besides eradicating introduced species.

[1 mark]

10

Discuss the reasons why the conservation of biodiversity is important for human society.

[9 marks]

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Extra space

10 . 2 State what is meant by **biodiversity**.

[1 mark]

10

Do not write outside the box

Write an essay on one of the following topics.

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Agriculture involves the control of abiotic and biotic factors to maximise food production. Discuss how this control can be achieved and evaluate its environmental impacts.

[25 marks]

OR

1 1 . 2

Discuss the methods that can be used to conserve biodiversity and evaluate their effectiveness.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

END OF QUESTIONS